NATIONAL UNIVERSITY



Third Year Syllabus

Department of Chemistry

Four-Year B.Sc. (Honours) Course
Effective from the Session: 2013–2014

National University

Subject: Chemistry
Syllabus for Four Year B.Sc. Honours Course
Effective from the Session: 2013-2014

Year wise Paper, Marks and Credit distribution

THIRD YEAR

Paper Code	Paper Title	Marks	Credits
232801	Physical Chemistry-III	100	4
232803	Stereochemistry	100	4
232805	Advanced Concepts of Atomic Structure and Chemical Bonding	100	4
232807	Coordination Chemistry and Organometallic Chemistry	100	4
232809	Fundamentals of Analytical Chemistry	100	4
232811	Industrial Chemistry	100	4
232813	Agricultural Chemistry	100	4
232814	Physical Chemistry Practical	50	2
232816	Organic Chemistry Practical (Detection & Preparation)	50	2
	Total =	800	32

Detailed Syllabus

Paper Code	232801	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title	**Physical Chemistry-III**			

1. **Chemical Kinetics**: Review of elementary concepts, integration of rate equations for model reaction systems: zero, first and 2nd order reactions. Parallel, consecutive, opposing reactions: methods for determination of order and rate constants, conventional method, conductomatric method, polarometer spectrophometer.

2. **Theoris of Reaction Rates:** Collision theory its success and failures. Elementary treatment of Transition State Theory.

3. **Catalysis:** Homogeneous and heterogeneous catalysis, acid-base catalysis; Hinshelwood and Rideal mechanism; enzyme catalysis; Michael-Menten equation.

4. **Reactions in Solution:** Influence of dielectric constant and ionic strength of medium on the rate constant of reactions, primary salt effect and secondary salt effect.

5. **Photochemistry and Radiation Chemistry:** Deference between photochemistry and radiation chemistry. Laws of photochemistry. Quantum yield and its significance. Some typical photochemical reactions. Formation and depletion of ozone in the stratosphere

6. **Colloidal State of matter:** Classification, preparation and physical properties, Electrokinetic phenomena; Collodial electrolytes and their uses, Emulsion; preparation, properties, stability and use.

7. **Surface Chemistry:** Solid surfaces and their characterization; Adsorption on solid surfaces: technique for measurement of adsorption from gas phase and solution; Langmuir, Frundlich and BET adsorption isotherm: Enthalpy of adsorption; Adsorption on liquid surface. Gibb's adsorption equation; Surface film; Electro-capillary phenomena.

8. **Phase Equilibrium:** Phase, components and degrees of freedom; phase rule; application in one component system like water and sulfur, Completely and partially miscible liquid pairs; Duhem-Margules equation. Two component system; solid-liquid system, solid-solid binary systems with reference to alloy, cooling curves, systems with and without compound formation, congruent and incongruent melting points; Efflorescence and deliquescence, vapour pressure of a saturated solution.

Books Recommended:

1. An Introduction to Chemisorption and Catalysis by Metals-R.P.A. Gasser.
2. Kinetics of Chemical Change, C. N. Hinshelwood.
3. Chemical Kinetics, K. J. Laidler.
4. Chemical Kinetics and Dynamics, J. I. Steinfield, J. F. Francisco and W. L. Hase. Prentic

Hall, New Jersy (1989).

5. Chemical Kinetics, K. A. Connors.
6. Chemical Kinetics and Reaction Mechanism, J. H. Espenson.

7. Introduction to Molecular Photochemistry, C. H. J. Wells.
8. Principles and Applications of Photochemistry, R. P. Wayne, Oxford. (1988).
9. Fundamentals of Photochemistry, K. K. Rohatgi and Mukhargi.
10. Phase Rule-Findlay. Revised by Compbell.
11. Physical Chemistry, P.W. Atkins, Oxford.

Paper Code	232803	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title	**Stereochemistry**			

1. **Fundamentals of Stereochemistry:** Stereochemistry and stereoisomerism.

2. **Isomerism:** *Cis-trans* isomerism, interconversion of *cis-trans* isomers, determination of configuration of geometrical isomers by chemical and physical methods, conformation of acyclic and cyclic molecules.

3. **Configuration**: D & L, threo erythro, R & S configurations. absolute and relative configurations and their correlation. relative and absolute configurations, determination of R-S configuration and their chemical interconversion; chiral, prochiral and pseudochiral molecules. chirality in molecules devoid of chiral centre; atropisomerism, biphenyl, allenes and spiranes.

4. **Optical Activity and Optical Isomerism:** Cause of optical activity. chirality (dissymmetry and asymmetry), prochirality, pseudo chirality. symmetry elements. optical isomers, diasterecoisomers, enantiomers, epimeners, anomers, meso and racemic compounds. racemic modification and their resolution.

5. **Geometrical Isomerism:** Conditions, configurations of geometrical isomers: *E/Z* system, syn-anti. physical properties and configurational ssignment of geometrical isomers. geometrical isomerism of polyenes, carbon-nitrogen, nitrogen-nitiogen double bonds.

6. **Stereochemistry of Fused Ring:** Stereochemistry of decalines, perhydroanthrancene, perhydrophenanthrene, their optical activity and relative stability.

7. **Conformation:** Conformation and conformers. conformation of propane, butane, ethanediol, dihydroxystyrene, dichlorostyrene, their physical properties and stability. stability of threo and crythro, meso and d1 compounds, conformation of cyclobutane, cyclopentane and cyclohexane and their stability.

8. **Optical Rotation and Rotatory Power:** Factors leading to chirality. molecular dissymmetry. atomic dissymmetry and conformational asymmetry.

Books Recommended:

1. I.L. Eliel, Stereochemistry of Carbon Compounds, Tata Mc Graw-Hill, 1979
2. R.T. Morrison and R.N. Boyd, Organic Chemistry, Fifth edition, Prentice-Hall of India, New Delhi, 1989.
3. Handrickson and Pyne, Organic Chemistry, McGraw-Hill.
4. I.L. Finar, Organic Chemistry Vol. II, ELBS, Fifty edication, 1985

Paper Code	232805	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title	**Advanced Concepts of Atomic Structure And Chemical Bonding**			

1. **Atomic Structure:** Wave-Particle duality, de Broglie's hypothesis for matter waves and its consequences, standing waves and quantization, Heisenberg's uncertainty principle, significance of the uncertainty principle, Schrödinger's wave equation and its application to hydrogen atom, solutions of Schrödinger wave equation, quantum numbers and their properties, radial wave functions, angular wave functions and shapes of the orbitals, probability distribution, radial and angular nodes, many electron atoms, sequence of energy levels, electronic configuration of atoms.

2. **Chemical Bonds:** Chemical bond, types of chemical bonds,

2.1 Ionic Bond: Ionic bond and its uniqueness, properties of ionic compounds, close packing, close-packed structures, interstitial sites, radius ratio, radius ratio rule, coordination number and radius ratio value, structure of NaCl, lattice energy of ionic crystals, theoretical calculation of lattice energy of NaCl crystal, Madelung constant, experimental determination of lattice energy of NaCl crystal, factors affecting the magnitude of lattice energy of ionic solids, applications of lattice energy calculation.

2.2 Covalent Bond: Covalent bond, properties of covalent bond and covalent bonded compounds, overlap integral and bond strength, polarity of bonds, dipole moment and percentage ionic character, factors affecting the magnitude of dipole moment, valence bond theory, hybridization, mathematical formulation of sp and sp2 hybrid orbitals, limitations of valence bond method, molecular orbital theory, the LCAO method, molecular orbital diagram, frontier orbitals - HOMO and LUMO, mixing of MOs and the correlation diagrams, MO descriptions of heteronuclear diatoms, HF and CO, Walsh diagram, MO descriptions of polyatoms like H2O, BeH2, and C6H6 involving π bonding electrons, comparison and contrast between VBT and MOT, applications of frontier orbital concept.

2.3 Bonding in Metals: Metallic bond, factors favoring the formation of metallic bond, theories of metallic bond - electron sea theory, molecular orbital theory, characteristics of metals, conductors, semiconductors and insulators.

2.4 Hydrogen Bond: Hydrogen bond, types of hydrogen bond, theories of formation of hydrogen bond – electrostatic approach, molecular orbital approach, properties of hydrogen bond and hydrogen bonded compounds, hydrogen bond in biological systems.

Books Recommended:

1. Chemical Structure and Bonding, R. L. DeKock and H. B. Gray, University Science Books.
2. Inorganic Chemistry, G. L. Miessler and D. A. Tarr, Prentice-Hall of India Private Limited.

3. Atomic Structure and the Chemical Bond, M. Chanda, Tata McGraw-Hill.
4. Physical Chemistry, G. M. Barrow, McGraw-Hill.
5. Inorganic Chemistry, D. F. Shriver, P. W. Atkins and C. H. Langford, Oxford University Press.
6. Valence and Molecular Structure, E. Cartmel and G. W. A. Fowles, Butterworths.

Paper Code	232807	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title	**Coordination Chemistry and Organometallic Chemistry**			

1. **Transition and Inner Transition Elements:** Transition elements, main and inner transition elements, their positions in the periodic table, main transition elements: general characteristics, melting and boiling points, variable oxidation states, color of complex ions, magnetism - origin of paramagnetism and diamagnetism, magnetic susceptibility, Curie's law, *Curie-Weiss law,* techniques of magnetic measurements, Gouy balance, ferromagnetism and antiferromagnetism, lanthanides and actinides.

2. **Concept of Coordination compounds:** Coordination compounds, double salts and coordination compounds, coordination sphere, coordination number, ligand types, properties of ligands, nomenclature of coordination compounds, isomerism in coordination compounds – structural isomerism and stereo isomerism, stability.

3. **Bonding in Coordination Compounds:** Coordination bond, Werner's coordination postulate, limitations of Werner's postulate, Sidgwick's electronic concept, application of EAN rule, limitations of Sidwick's concept, assumptions of valence bond theory (VBT), hybridization and geometry of complexes, inner orbital and outer orbital octahedral complexes, limitations of VBT, important features of crystal field theory (CFT), orbital splitting and electron spin, factors influencing the magnitude of 10dq, spectrochemical series, high and low spin complexes, crystal field stabilizing energies of dn configuration (n = 0 to 10), four and six coordination preferences, magnetic moments, distortion of octahedral complexes and Jahn-Teller theorem, limitations of CFT, ligand field theory (LFT).

4. **Reactions and Mechanisms in Coordination Chemistry:** Inert and labile ligands, substitution reactions, types of substitution reactions, nucleophilic substitution reactions, association, dissociation and interchange mechanisms, factors affecting the rate of substitution reactions, acid and base hydrolysis reactions, the conjugate base mechanism, stereochemistry of octahedral substitution, substitution in square planar complexes, trans effect – theories of trans effect, uses of trans effect, substitution in tetrahedral complexes, fluxionality in coordination compounds.

5. **Metal Carbonyls and Nitrosyls:** Pi-acid ligands and pi-acid complexes, metal carbonyl and metal nitrosyl complexes, preparation and properties of metal carbonyl and nitrosyl

complexes, M-C-O and M-N-O bonding, bridging and terminal COs and NOs, infrared and 13C NMR analysis of carbonyls and nitrosyls, biological role of nitrosyl compound.

6. **Organometallics and Their Catalytic Aspects:** Introduction, general characteristics, ligands in organometallic chemistry, 18- and 16-electron rule, stability of organometallic compounds, classification of organometallic compounds, preparative routes for metal-carbon bond formation, bonding between metal atoms and organic pi systems, structures of Zeise's salt and ferrocene, organometallic reactions – ligand dissociation and substitution, oxidative addition, reductive elimination, carbonyl insertion, homogeneous catalysis, hydrogenation by Wilkinson's catalyst, hydroformylation, heterogeneous catalysis, Ziegler-Nata polymerizations, water gas reactions, Fisher-Tropsch reaction.

Books Recommended:

1. Inorganic Chemistry, J. E. Huheey, E. A. Keiter, and R. L. Keiter, Harper Collins College Publishers.
2. Advanced Inorganic Chemistry, F. A. Cotton, G. Wilkinson, C. A. Murillo and M. Bochmann, John Wiley & Sons, Inc.
3. Selected Topics on Advanced Inorganic Chemistry, S. Z. Haider, Students' Publications.
4. Inorganic Chemistry, D. F. Shriver, P. W. Atkins and C. H. Langford, Oxford University Press.
5. Kinetics and Mechanism, A. A. Frost and R. G. Pearson, John Wiley & Sons.
6. Inorganic Reaction Mechanisms, M. L. Tobe, Thomas Nelson and Sons Ltd.

Paper Code	232809	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title	**Fundamentals of Analytical Chemistry**			

1. Basic Concepts in Analytical Chemistry and Statistical Treatment of Data: **Analytical detection and quantification, sensitivity, selectivity, specificity, concentration limit, dilution limit etc. of chemical reactions, sample containers, sample preservation, sampling, sample dissolution, wet ashing and dry ashing, reagents and reactions, population and sample mean, standard deviation, relative standard deviation, coefficient of variation, variance, confidence limit, Gaussian distribution, statistical tests – the F test, the T test, the Q test, regression lines, least square method, coefficient of correlation.**

2. Volumetric Analysis: **Principle, apparatus, end point, indicator, general factors influencing volumetric method, advantages, acid-base titrations, redox titrations, complexometric titrations – complexing agents,** influence of [H+] on complexation

3. **Gravimetric Analysis:** Introduction, general requirements of a gravimetric method, precipitation from homogeneous solution.

4. **Spectrophotometric Analysis:** Ultraviolet and visible radiation, absorbance, transmittance, absorptivity, the Beer-Lambert's law, limitations of Beer-Lambert's law, basic components of a spectrophotometer, qualitative and quantitative analysis.

5. **Thermal Analysis:** Thermogravimetry (TG), types of TG, instrumentation, application of TG, differential thermal analysis (DTA): working principle, instrumentation, factors affecting DTA, applications, differential scanning calorimetry (DSC): working principle, instrumentation and applications.

6. **Atomic Spectrometric Methods:** Atomic absorption and atomic emission, absorption line width, choice of absorption line, flame emission spectrometry: instrumentation, flame emission analysis, atomic absorption spectrophotometry: principles, instrumentation and interferences, hydride vapour generation technique, cold vapour technique, advantages and disadvantages of AAS.

7. **Polarographic and Voltammetric Analysis:** Current voltage relationship, mass transport processes, direct current polarography (DC), diffusion current, charging current, factrors affecting the diffusion current, characteristics of dropping mercury electrode, three electrode potentiostat, polarographic maxima, oxygen interference, half wave potential, alternating current and pulse polarography, principle and advantages over dc polarography, voltammetry- ASV, CSV and CV, multicomponent analysis, quantitative applications.

Books Recommended:

1. Fundamentals of Analytical Chemistry, D. A. Skoog, D. M. West, F. J. Holler and S. R. Crouch, Saunders College Publishing.
2. Analytical Chemistry, G. D. Christian, John Wiley & Sons.
3. Modern Analytical Chemistry, D. Harvey, McGraw-Hill Higher Education.
4. A Text Book of Quantitative Analysis, A. I. Vogel, Longman, Green and Co. Ltd.

Paper Code	232811	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title	**Industrial Chemistry**			

1. **Fundamentals in the Development of Chemical Industries:** General ideas about unit processes and unit operations, raw materials, process design, commercial energy sources, catalysts, heat transfer, mass transfer, separation processes, concepts of consumption, production, and market evaluation, safety, environmental considerations, site and technology selection criteria, cost-benefit analysis.

2. **Chlor-alkali Industries:** Raw materials, manufacture of caustic soda, soda ash, sodium bicarbonate, chlorine, bleaching powder, environmental hazards of these chemicals.

3. **Cement Industries:** Portland cement, raw materials, important process parameters for manufacturing a good cement clinker, methods of manufacturing Portland cement, sequence of operations, additives for cement, properties of cement, testing of cement, setting of cement, other types of cement.

4. **Glass Industries:** Properties of glass, raw materials and fundamentals of glass industries, methods of manufacture, choice of the furnace, chemical reactions in the furnace, annealing, special glasses.

5. **Ceramic Industries:** Ceramics, properties of ceramics, basic raw materials, manufactures of ceramics, refractories, requisites of a good refractory, classification of refractories, properties of refractories, manufacture of refractories, types of refractory products.

6. **Acids:** Raw materials, manufacture of hydrochloric, phosphoric, sulfuric acids, concentration and purification of acids, industrial uses of mineral acids, safety and hazards.

7. **Iron and Steel Industry:** Fundamentals of metallurgy, ores of iron, three commercial forms of iron, construction and operation of blast furnace, reactions in blast furnace, byproduct in blast furnace, classification of steel, steel manufacturing processes, effects of impurities on steel, phases in Fe-C system.

8. **Sugar and Starch Industries:** Steps in the industrial extraction of cane sugar, and inversion of sugar, refining of sugar, production of sugar from sugar beet, byproducts of sugar industries, management of industrial waste of sugar industries, production of starch from corn, production of glucose and dextrin from starch.

9. **Cellulose and Allied Industries:** Natural sources of cellulose, its constituents and their estimation, different processes for the manufacturer of paper from pulp, production of paper board, viscose rayon and other modified cellulosic fiber, wood chemistry and wood chemicals.

10. **Fuels:** Solid, liquid and gaseous fuels, coal and its constituents, different stages of coal formation, analysis and calorific value of coal and other fuels, manufacture of producer gas and water gas, refining and distillation of crude oil, thermal and catalytic cracking, hydrocarbons and petroleum, petroleum refining, petrochemicals from liquid and gaseous hydrocarbons, natural gas and its utilization.

11. **Natural Oils, Fats and Waxes:** Extraction and refining of vegetable oils, analysis of fats and oils and their uses, hydrogenation of oils - soybean and sunflower oils and their uses.

12. Paints and Varnishes: Constituents of paints, varnishes, lacquers, enamels and printing inks and their functions, paints and pigments and their manufacturer, differentiation between paints and varnishes.

13. Soaps and Detergents: Methods of fat splitting, manufacturer of laundry and toilet soaps, recovery and refining of glycerin, detergents: definition, classification and their manufacture, comparison between soaps and detergent.

Books recommended:

1. Chemical Process Industries, R. N. Shreve and J. A. Brink, Jr., McGraw-Hill Inc.
2. Industrial Chemistry, B. K. Sharma, Geol Publishing House.
3. Reagel's Hand Book of Industrial Chemistry, J. A. Kent edited, Van Nostrand.
4. Chemical Process Industries, G. T. Austin edited, McGraw-Hill.
5. Materials Science and Engineering – An Introduction, W. D. Callister, Jr., John Wiley & Sons, Inc.
6. Inorganic Medicinal and Pharmaceutical Chemistry, J. H. Block and E. B. Roche, Lea and Febiger Pub.
7. Environmental Chemistry Vol. I - III, H. J. M. Bowen, The Royal Society of Chemistry.

Paper Code	232813	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title	**Agricultural Chemistry**			

1. **Pesticides:** Its importance, Classification with examples

2. Important Insecticides, Acaricides nematocides, Moluscicides, Rodenticides, Herbicides, Fungicides etc.

3. **Toxicity of Pesticides:** Lethaldose, Toxicsurblethaldose, LD50, ED50, EC50 etc.

4. **Effects of pesticides:** Acute effect, chronic effect, systemic effects etc.

5. **Formulation Pesticides:** Grannular, Wettable powder, Emulsion etc.

6. **Metabolism** of pesticides in biological system, detoxification of pesticides and their metabolites in the environment

7. **A brief introduction of organochlorine compounds** as pesticides with reference to lindane, heptachlor., DDT, endosulphane etc. The mechanism of action of organochlorine compounds in the biological system, detoxification of the organochlorine and the residual effects in the environment.

8. **A brief introduction of organophosphorous compounds** and carbamates and their mechanism of actions in the biological system, their metabolism and cause of resistance in the insect system.

9. **Studies of a few pesticides:** Metaphos, Bromophos, Metathion, Diazinon, Melathion, Carbaryl, Pirimicarb etc.

10. Fungicides, herbicides, rodenticides and their classifications.

11. Dithiocarbamic acids derivatives, zineb, captan, benomyl, pyrazophos, hexachlorobenzene (HCB), phenol derivatives, 2,3,6-TBA,2,4-D, MCPA, MCPB, 2,4-DB etc.

12. Mechanism of action of 2,4-D derivatives.

13. Insect attractant, repellant, chemosterile retardants etc.

14. **Natural Pesticides:** Pest control by compounds from natural products. Pyrenthrines, rotenone, glucosoinolates, azadirectin etc.

15. **Pest control** by pheromones. Advantage of using pheromones over synthetic pesticides.

16. **Fertilizer Industries:** Plant nutrients, classification of fertilizers, natural inorganic fertilizers, nitrogen fixation, artificial fertilizers, manufacture of ammonia and urea, action of urea as fertilizer, potassium fertilizer, calcium phosphate and other phosphatic fertilizers, NPK fertilizer.

Books Recommended:

1. G.S. Gruzdyev VA. Zinchenko, V.A. Kalinin, R.I. Slovisov, The Chemical Protection of Plants. Mir Publishers Moscow.

Paper Code	232814	**Marks: 50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title	**Physical Chemistry Practical (6 Hours Duration)**			

List of Experiments

1. A study of adsorption of acetic acid from solution on a sample of charcoal.

2. Determination of the surface area of a surface active molecule: n-butanol.

3. Determination of the rate constant for the alkaline hydrolysis of ethylacetate.

4. Determination of the association constant for the dimerization of benzoic acid in toluene by measuring distribution between water and toluene.

5. Determination of dissociation constant of acids by conductometric technique.

6. Conductometric titration (i) strong acid-strong base, (ii) weak acid-strong base and (iii) mixture of weak and strong acids-strong base.

7. Determination of dissociation constant of mono basic and dibasic acids by pH titration.

8. Study of electronic spectra of I2 in CH2Cl2 and toluene, verification of the Beer-Lambert law using KMnO4 solution.

9. Spectrophotometric determination of Iron: application of Beer-Lambert law.

Books Recommended:

1. Practical Physical Chemistry, A Faraday
2. Experimental Physical Chemistry, Deniel, Mathews and William
3. Findlay's Practical Physical Chemistry, Revised by BP Levitt
4. Experiments in Physical Chemistry, D.P Shomeker and C.W. Garland
5. Experiments of Physical Chemistry, F. Deniels, J.N. William, P.Bender, R.A. Alberty, C.D. Cornwell, and J.E. Harriman
6. Chemistry Experiments for Instrumental Methods. D.T. Sawer, W.R. Heinman and J.M. Beebe.

Paper Code	232816	**Marks: 50**	**Credits: 2**	**Class Hours: 30 hrs.**
Paper Title	**Organic Chemistry Practical (6 Hours Duration)**			

1. **Systematic Identification of Organic Compounds:** Determination of melting temperature, Tm, boiling temperature, Tb; detection of N, S and X in organic solids and liquids, determination of functional groups; naming of compounds.
2. **Organic Synthesis:** Preparation of acetanilide, 4-nitroacetanilide, 4- nitroaniline, 4-iodoaniline, benzoic acid, 4-bromoacetanilide and Methyl Orange.

Books Recommended:

1. Qualitative Organic Analysis – A.I Vogel
2. A Text Book of Practical Organic Chemistry – Clarke
3. Identification of Organic Compounds – Shiriner and Fuson.